

HCP Profiling App • Bug Inventory & Release Notes Guide

Complete Technical Catalog of Encountered Defects, Root Causes, Architectural Fixes, and Documentation Standards

Active Version: V.0.4.8 (Build 19) **Target Backend:** ERPNext v15 Multi-Tenant **Scope:** SFE, MedRep & DSM Workflows

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1. Executive Summary & Defect Taxonomy

Throughout the lifecycle of the **HCP Profiling App** (from foundational release V.0.1.0 up to current release V.0.4.8), defects encountered during field testing, sprint reviews, and ERPNext integration were resolved through systematic patch releases. Documenting these bugs in **Release Notes** is critical for audit trails, stakeholder visibility, and QA regression testing.

The 4 Primary Defect Domains Identified in the System:

- 1. Workflow & Role Permission Lockouts:** ERPNext workflow transition rejections (HTTP 417 / 403) resulting from mismatched DocType roles.
- 2. Deletion & Data Loss Vulnerabilities:** Accidental permanent deletion of proposed facilities via "Discard" options, preventing continuous resubmission.
- 3. UI Misclassification & Data Bleed:** Master facilities imported in Draft state being flagged as rejected, and red rejection banners leaking to approved profiles.
- 4. Backend Synchronization & Mock Traps:** Silent client-side mock fallbacks that masked genuine ERPNext validation errors.

2. Comprehensive Master Bug Inventory (V.0.1.0 to V.0.4.8)

VERSION	BUG TITLE & CONTEXT	ROOT CAUSE DISCOVERED	SOLUTION APPLIED & RELEASE NOTE INCLUSION	STATUS
V.0.4.8 +1 Patch	Dropdown Search Missing Prefix Prioritization & Clue Ingestion	Institutions starting with search query lacked explicit top-tier grouping; bidirectional query-shortening expansion was missing.	Prefix Prioritization & Clue Search: Added Tier 1 +2500.0 pts prefix bonus ensuring all facilities starting with query display at beginning; secondary A-Z sort; bidirectional token retention.	RESOLVED
V.0.4.7 +1 Patch	Overly Rigid Search Filterizing Out Valid Institution Matches	Rigid single-string contains() check failed on multi-token clues, out-of-order words, and medical abbreviations.	Fuzzy Multi-Clue Search Engine: Token-based fuzzy search with synonym/abbreviation expansions, acronym resolution, and inclusive clue retention.	RESOLVED
V.0.4.6 +1 Patch	Accidental Deletion of Added Institutions	MedRep "Discard" button and SFE "Reject & Delete" branching permanently deleted facilities, preventing iterative corrections.	Zero-Deletion Policy: Removed Discard button. Replaced with prominent "Modify & Resubmit". SFE rejections strictly set status to Rejected without deleting.	RESOLVED
V.0.4.5 +1 Patch	SFE Rejection Fails (HTTP 417 Error)	ERPNext Institution WF strictly required role Sales Force Effectiveness. SFE account	Workflow Role Alignment: Assigned SFE role to account; added System Manager & Sales	RESOLVED

VERSION	BUG TITLE & CONTEXT	ROOT CAUSE DISCOVERED	SOLUTION APPLIED & RELEASE NOTE INCLUSION	STATUS
		lacked this role, causing false "check internet" toast.	Manager to workflow transitions; added CSRF check and direct PUT fallback.	
V.0.4.4 +1 Patch	Misleading Rejection Remarks on Approved Profiles	General Frappe timeline comments were mapped to rejectionRemarks without verifying if document was actually rejected.	Rejection Remarks Isolation: Enforced strict gating (docstatus == 2 workflow_state == 'Rejected') before displaying red banners.	RESOLVED
V.0.4.3 +1 Patch	Pending Facilities Blocked from Doctor Profiling	Validation logic in wizard blocked MedReps from selecting facilities in Pending Approval or Draft, halting field visits.	Unlocked Pending Profiling: Allowed pending facilities for doctor profiling; added dynamic notes ("this institution is not yet approved" → "this institution is now approved").	RESOLVED
V.0.4.2 +1 Patch	3,900+ Master Facilities Misclassified as Rejected	Mobile app treated all non-approved institutions as rejected, locking out 3,900+ baseline national registry facilities imported as Draft.	ERPNext Status Color Parity: Aligned status logic with ERPNext Desk (Draft badged in Red pill, matching ERPNext) and restored profiling linkage across all 3,990+ facilities.	RESOLVED
V.0.4.1 +1 Patch	Fragmented Directory & Screen Space Clutter	Directory was split across status tabs; uncollapsible search filter row consumed over 40% of vertical screen space on mobile phones.	Unified Directory & Filter Bar: Merged into single masterlist; added animated collapsible [= Filter] bar with dynamic record counter.	RESOLVED
V.0.2.6 +1 Patch	Unapproved Clinics Leaking into Doctor Profiles	Inverted boolean check in _hasUnapprovedWorkplaces allowed unapproved custom clinics to register in dynamic lookup before SFE approval.	Strict Gating Enforcement: Hardened isApprovedForProfiling across wizard pickers, search dialogs, and backend API submission payloads.	RESOLVED
V.0.2.4 +1 Patch	MedRep "Discard" Not Deleting on Server	ERPNext Custom DocPerm for Sales User had delete: 0 (HTTP 403 Forbidden). App silently cleared cache without server confirmation.	Permission & CSRF Fix: Granted delete: 1 on ERPNext for Sales User; added ensureCsrfToken(); removed premature cache-purge.	RESOLVED
V.0.2.3 +1 Patch	Deceptive "Faking Submission" on Server Error	When ERPNext returned HTTP 417 LinkValidationError on location fields, the app caught the error and faked success in local cache without server persistence.	Eliminated Mock Fallbacks: Removed mock fallback in api_service.dart; implemented 10-digit PSGC auto-fallback; surfaced genuine error messages.	RESOLVED
V.0.2.1 +1 Patch	New Facilities Not Appearing in SFE Queue	App attempted to insert records directly as Pending Approval, violating Frappe workflow rules requiring initial insertion in Draft.	Two-Stage Insert Flow: Created institution in Draft state first, then immediately triggered the Submit for Approval transition.	RESOLVED
V.0.1.0 Baseline	Doctor Duplication & Denominator Miscalculations	Editing doctors created duplicate submissions; submission denominator counts were misaligned across roles; raw PSGC codes displayed instead of names.	Core Architecture Fixes: Two-Tier Masterlist (HCP Universal vs Program HCP Account); in-place updates; human-readable PSGC translation dictionary.	RESOLVED

3. Technical Post-Mortems of Major Bug Resolutions

1. Institution Zero-Deletion Policy (V.0.4.6)

RELEASE V.0.4.6

Problem Statement: MedReps had a "Discard & Delete" option that permanently deleted rejected institutions. If discarded or rejected a second time, proposals were expunged from the database, eliminating audit history and forcing reps to start from scratch.

Technical Cause: `_handleDeleteSubmission()` in `InstitutionApprovalsScreen` and `deleteInstitution()` in `ApiService` called `DELETE /api/resource/Institution/...`

Solution Applied: Completely removed all delete methods and discard buttons from both MedRep and SFE screens. Replaced with prominent "**Modify & Resubmit**" button allowing continuous iterative edits. SFE rejection now strictly marks state as Rejected without deletion.

2. SFE Workflow Transition Lockout HTTP 417 (V.0.4.5)

RELEASE V.0.4.5

Problem Statement: When the SFE Specialist clicked "Reject" on a facility proposal, the app displayed a generic toast: "*Please verify internet connection*". The rejection remarks were saved, but the facility status remained stuck at Pending Approval.

Technical Cause: ERPNext's Institution WF workflow transition strictly mandated the role Sales Force Effectiveness. The active user account possessed System Manager and Sales Manager, causing ERPNext to reject the transition with HTTP 417: Not a valid Workflow Action.

Solution Applied: Assigned Sales Force Effectiveness role to active SFE account; expanded Institution WF transition permissions to include System Manager and Sales Manager; added CSRF tokens and a direct PUT fallback in `ApiService.rejectInstitution`.

3. Rejection Remarks Display Bleed (V.0.4.4)

RELEASE V.0.4.4

Problem Statement: Approved and processed doctor profile submissions were displaying alarming red rejection remark banners in both list tiles and detail drawers.

Technical Cause: The model deserializer (`HcpProfileSubmission.fromJson`) pulled the most recent Frappe timeline comment and assigned it to `rejectionRemarks` without checking if the submission was actually in a rejected state.

Solution Applied: Guarded `rejectionRemarks` mapping with an explicit `isActuallyRejected` check (`rawDocstatus == 2 || resolvedWorkflow == 'Rejected'`). UI cards strictly check `item.isRejected` && `item.rejectionRemarks != null` before rendering red banners.

4. Pending Facility Profiling Lockout (V.0.4.3)

RELEASE V.0.4.3

Problem Statement: Medical representatives could not complete doctor profiling if their proposed clinic was awaiting SFE approval, causing severe profiling bottlenecks during field visits.

Technical Cause: Hardcoded checks in `HcpWizardScreen._showAddWorkplaceSelector` blocked any institution with `workflow_state != 'Approved'`.

Solution Applied: Unlocked Pending Approval facilities for doctor profiling. Implemented dynamic approval notes: "*this institution is not yet approved*" (amber hourglass) which flips to "*this institution is now approved*" (emerald green check) upon SFE approval.

5. 3,900+ Master Facilities Misclassification (V.0.4.2)

RELEASE V.0.4.2

Problem Statement: Doctors could not be associated with over 3,900 valid hospitals because the mobile application displayed them with a red rejected badge.

Technical Cause: Initial facilities imported into ERPNext have `workflow_state: 'Draft'`. The mobile app treated anything non-approved as rejected.

Solution Applied: Aligned workflow status logic directly with ERPNext Desk. Draft masterlist facilities are badged in red Draft pill and are fully selectable for doctor profiling.

4. Standard Operating Procedure: Writing Bug Fixes in Release Notes

To maintain professional release notes in `RELEASE_NOTES.md` and `CHANGELOG.md`, every bug fix must strictly adhere to the following 4-part structure:

```
### 🐛 Bug Fix: [Component Name / Feature Title] (+1 Patch)

- **Problem Encountered:**
  - Concise description of user-facing failure, error toast, or workflow blockage.
- **Root Cause Identified:**
  - Technical analysis (e.g., missing workflow transition role, invalid state mapping, API permission).
- **Solution Applied:**
  - Code changes made in Flutter mobile app and/or configuration updates on ERPNext backend.
- **Verification & Impact:**
  - Test scenario executed, status transition confirmed, and release APK binary tagged.
```